

Sharp logarithmic corrections under stretched exponential target bounds

Supplementary mathematical note 7 October 2026

Abstract

The stretched-exponential upper bound in manuscript 007 has logarithmic power $2/(3q)$ at Wasserstein exponent $1/3$. Reweighting the explicit two-atom constructions of manuscript 001 v3 shows that this logarithmic power is necessary for a fixed uniform cube source and for a fixed cube-truncated Gaussian source. The latter is strongly log-concave with a hard convex boundary. In contrast, the full Gaussian source has a pure one-third bound on the same target class. We give the complete lower-bound calculation, including the source-cell masses, direct optimality of the maps, optimal target coupling, and logarithmic asymptotics.

Scope and dependencies. This note combines existing proved parts of two research drafts. The two-atom geometries are those of manuscript 001 v3, Theorem 1.7, Section 11 and Section 12.2 [1]. The upper estimate is manuscript 007 v1, Corollary 4.1, using its Theorem 3.1 and Corollary 3.2 [2]. The lower bounds below are proved directly and do not use the radius-independent potential-stability theorem. The source-specific upper bounds use the potential theorems proved in manuscript 001 v3, Theorems 1.1 and 1.4; this note does not independently revalidate those theorems. No novelty, publication-priority, external-peer-review, or journal-level claim is made. Neither frozen manuscript is changed by this note.

1 Statement and common calculation

For $q, K > 0$, define the pair class

$$\mathcal{E}_{q,K} = \left\{ (\mu, \nu) : \int e^{(|y|/K)^q} d\mu(y) + \int e^{(|y|/K)^q} d\nu(y) \leq 4 \right\}.$$

Every member has finite moments of every positive order. For an absolutely continuous source ρ , let T_μ be its quadratic Brenier map to μ , and write

$$w = W_2(\mu, \nu), \quad X_\rho(\mu, \nu) = \|T_\mu - T_\nu\|_{L^2(\rho)},$$

$$\Phi_{q,K}(w) = K^{2/3} w^{1/3} (1 + \log(K/w))^{2/(3q)}, \quad 0 < w \leq K.$$

Theorem 1. Fix $d \geq 2$ and either of the following sources, independently of q, K and of the target parameter:

1. the uniform probability measure on $[-1, 1]^d$;
2. the standard Gaussian conditioned on $[-1, 1]^d$.

For every $q, K > 0$, there are pairs $(\mu_a, \nu_a) \in \mathcal{E}_{q,K}$, each target having exactly two atoms, such that $w_a \rightarrow 0$ as $a \downarrow 0$, and

$$\frac{X_\rho(\mu_a, \nu_a)}{\Phi_{q,K}(w_a)} \longrightarrow c_{\rho,q} \in (0, \infty).$$

In particular, no uniform estimate with Wasserstein power $1/3$ and logarithmic power $\gamma < 2/(3q)$ holds on this class. Nor can $\Phi_{q,K}$ be replaced by a function that is $o(\Phi_{q,K})$ as $w \downarrow 0$.

For these sources, manuscript 007, Corollary 4.1, gives the matching upper bound $X_\rho \leq C_{\rho,d,q} \Phi_{q,K}(w)$ for $(\mu, \nu) \in \mathcal{E}_{q,K}$ and $0 < w \leq K$, using the explicitly cited potential theorems proved in manuscript 001.

The second source has density

$$r(x) = Z_d^{-1} e^{-|x|^2/2} \mathbf{1}_{[-1,1]^d}(x).$$

It is 1-strongly log-concave in the extended-potential sense: $r = Z_d^{-1} e^{-|x|^2/2-W}$ with the proper lower-semicontinuous convex function $W = I_{[-1,1]^d}$, equal to zero on the cube and $+\infty$ outside. Consequently, the necessity of the logarithm holds even for one fixed strongly log-concave source with a hard boundary.

Lemma 2 (Equal-mass two-atom formulas). *Let $0 < m < 1/2$, $R > 0$, and let e, n be unit vectors with $e \cdot n > 0$. Put $\eta = |e - n|$ and*

$$\mu = (1 - m)\delta_0 + m\delta_{Re}, \quad \nu = (1 - m)\delta_0 + m\delta_{Rn}.$$

Then

$$W_2(\mu, \nu)^2 = mR^2\eta^2. \tag{1}$$

If their Brenier maps from ρ have the form $T_\mu = Re\mathbf{1}_A$ and $T_\nu = Rn\mathbf{1}_B$, where $\rho(A) = \rho(B) = m$, and $D = \rho(A \triangle B)$, then

$$X_\rho(\mu, \nu)^2 = R^2[D + \eta^2(m - D/2)]. \tag{2}$$

Finally, taking $R = K[\log(1/m)]^{1/q}$ puts the pair in $\mathcal{E}_{q,K}$, with combined exponential moment exactly $4 - 2m$.

Proof. A coupling is determined by the mass $s \in [0, m]$ sent from Re to Rn . Its other three entries, in the natural two-by-two table, are $m - s$, $m - s$, and $1 - 2m + s$. Its cost is

$$R^2[2(m - s) + s\eta^2] = 2R^2[m - s(e \cdot n)].$$

Since $e \cdot n > 0$, this is minimized at $s = m$, proving (1). The map discrepancy has squared magnitude R^2 on $A \triangle B$, $R^2\eta^2$ on $A \cap B$, and zero elsewhere. As $\rho(A \cap B) = m - D/2$, this proves (2). For either target, the exponential moment is

$$1 - m + m \exp((R/K)^q) = 1 - m + m \exp(\log(1/m)) = 2 - m.$$

□

Direct optimality of the ramp maps. Both constructions below use finite convex functions that are maxima of two affine functions, with slopes exactly the two desired target atoms. To verify optimality without invoking an imported stability result, write such a function as $u(x) = \max_{y \in \{0, y_1\}} (x \cdot y + c_y)$. Then

$$x \cdot y + c_y \leq u(x)$$

for both target atoms, with equality for the selected gradient slope. Integrating against any coupling to the same target bounds $\int x \cdot y$ by the fixed number $\int u d\rho - \sum_y c_y \mu(\{y\})$. The gradient coupling attains this number. Since both marginal second moments are fixed, it minimizes quadratic cost. The separating hyperplane is source-null, so equality selects the unique gradient slope almost everywhere, also proving uniqueness of the optimal map. All integrals here are finite, because each source is compact and each individual target is finite.

2 Exact uniform-cube construction

Let ρ be the uniform probability on $[-1, 1]^d$. For $0 < a < 1/2$, set

$$m = \frac{a}{2}, \quad n_a = \frac{e_1 + ae_2}{\sqrt{1+a^2}}, \quad \eta_a = |e_1 - n_a|, \quad R_a = K[\log(2/a)]^{1/q}.$$

Define the rare source cells

$$A_a = \{x_1 > 1 - a\}, \quad B_a = \{x_1 + ax_2 > 1 - a\}.$$

Conditional on $x_2 = z \in [-1, 1]$, their horizontal widths inside the cube are a and $a(1+z)$. Both are in $[0, 2]$. The two intervals share their right endpoint and hence are nested. If Z is uniform on $[-1, 1]$, their source probabilities are therefore exactly

$$\rho(A_a) = \rho(B_a) = a/2, \tag{3}$$

$$D_a = \rho(A_a \triangle B_a) = \frac{a}{2} \mathbb{E}|Z| = \frac{a}{4}, \tag{4}$$

$$\rho(A_a \cap B_a) = \frac{a}{2} \mathbb{E} \min(1, 1 + Z) = \frac{3a}{8}. \tag{5}$$

Here the factor $1/2$ in each horizontal probability is the normalized one-dimensional cube density. The remaining $d - 2$ coordinates integrate out.

The convex potentials

$$u_a(x) = R_a(x_1 - 1 + a)_+, \quad v_a(x) = \frac{R_a}{\sqrt{1+a^2}}(x_1 + ax_2 - 1 + a)_+$$

have gradients $R_a e_1 \mathbf{1}_{A_a}$ and $R_a n_a \mathbf{1}_{B_a}$. By (3), their laws are the two-atom targets in Lemma 2. The direct argument above proves that these are their Brenier maps. The lemma now gives the exact identities

$$w_a^2 = m R_a^2 \eta_a^2, \quad X_a^2 = m R_a^2 \left(\frac{1}{2} + \frac{3}{4} \eta_a^2 \right), \quad \eta_a^2 = 2 - \frac{2}{\sqrt{1+a^2}} \sim a^2. \tag{6}$$

In particular, the combined exponential moment is $4 - a \leq 4$.

Put $\ell_a = \log(2/a)$. Since every fixed power of ℓ_a grows more slowly than every inverse positive power of a ,

$$w_a \sim \frac{K}{\sqrt{2}} a^{3/2} \ell_a^{1/q} \longrightarrow 0, \quad X_a \sim \frac{K}{2} a^{1/2} \ell_a^{1/q}.$$

Moreover

$$\log(K/w_a) = \frac{3}{2} \log(1/a) - \frac{1}{q} \log \ell_a + \frac{1}{2} \log 2 + o(1) \sim \frac{3}{2} \ell_a.$$

Consequently, for every real γ ,

$$\frac{X_a}{K^{2/3} w_a^{1/3} (1 + \log(K/w_a))^\gamma} \sim 2^{-5/6} (3/2)^{-\gamma} \ell_a^{2/(3q) - \gamma}. \quad (7)$$

This tends to infinity for $\gamma < 2/(3q)$ and to the positive constant $2^{-5/6} (2/3)^{2/(3q)}$ at equality. The uniform-cube part of Theorem 1 follows.

3 A fixed cube-truncated Gaussian

Let ρ be the standard Gaussian conditioned on $[-1, 1]^d$. Its density is the product of copies of

$$f(x) = Z_1^{-1} e^{-x^2/2} \mathbf{1}_{[-1, 1]}(x), \quad Z_1 = \int_{-1}^1 e^{-s^2/2} ds.$$

Write its first two coordinates as independent symmetric variables X, Z , and put $\ell = f(1) > 0$. For $0 \leq s \leq 2$, define

$$F(s) = \Pr(X > 1 - s) = \int_{1-s}^1 f(x) dx.$$

This function is continuous and strictly increasing. Smoothness of f up to the right endpoint from the interior gives

$$F(s) = \ell s + O(s^2) \quad (s \downarrow 0), \quad (8)$$

uniformly for $0 \leq s \leq Ca$ at every fixed C .

Fix once and for all $k = 1/8$. For small $a > 0$, put $b = ka$ and $m_a = F(a)$. Choose $c_a \in [(1 - k)a, (1 + k)a]$ by

$$\mathbb{E}F(c_a + bZ) = F(a). \quad (9)$$

Existence and uniqueness follow directly: throughout the candidate interval, $c + bZ$ lies in $[(1 - 2k)a, (1 + 2k)a] \subset (0, 2)$ for small a . At its lower endpoint the widths are at most a , strictly less almost surely, so the expectation is below $F(a)$; at the upper endpoint it is above $F(a)$. Continuity and strict monotonicity complete the argument. Since $\mathbb{E}Z = 0$, (8) yields

$$c_a = a + O(a^2), \quad m_a \sim \ell a. \quad (10)$$

Take

$$A_a = \{x_1 > 1 - a\}, \quad B_a = \{x_1 + bx_2 > 1 - c_a\}.$$

Their masses are exactly m_a by (9). On each $x_2 = z$ slice their intervals are nested and have probabilities $F(a)$ and $F(c_a + bz)$. Thus their symmetric-difference mass is exactly

$$D_a = \mathbb{E}|F(c_a + bZ) - F(a)|. \quad (11)$$

Equations (8)–(10) imply, uniformly for $z \in [-1, 1]$,

$$F(c_a + bz) - F(a) = \ell k a z + O(a^2).$$

The absolute-value function is 1-Lipschitz, so taking absolute values and expectations gives

$$D_a = \ell k \mathbb{E}|Z| a + O(a^2), \quad D_a/m_a \longrightarrow k \mathbb{E}|Z| > 0. \quad (12)$$

The overlap mass is exactly $m_a - D_a/2$.

Now choose

$$n_a = \frac{e_1 + be_2}{\sqrt{1+b^2}}, \quad \eta_a = |e_1 - n_a|, \quad R_a = K[\log(1/m_a)]^{1/q}.$$

The convex potentials

$$u_a(x) = R_a(x_1 - 1 + a)_+, \quad v_a(x) = \frac{R_a}{\sqrt{1+b^2}}(x_1 + bx_2 - 1 + c_a)_+$$

give the exact Brenier maps $R_a e_1 \mathbf{1}_{A_a}$ and $R_a n_a \mathbf{1}_{B_a}$, by the same direct supporting-affine-function argument. Their target laws are

$$\mu_a = (1 - m_a)\delta_0 + m_a\delta_{R_a e_1}, \quad \nu_a = (1 - m_a)\delta_0 + m_a\delta_{R_a n_a}.$$

For small a , $m_a < 1/2$ and $e_1 \cdot n_a > 0$. Lemma 2 proves that the pair has combined exponential moment $4 - 2m_a \leq 4$ and satisfies the exact formulas

$$w_a^2 = m_a R_a^2 \eta_a^2, \quad X_a^2 = R_a^2 [D_a + \eta_a^2 (m_a - D_a/2)]. \quad (13)$$

Also

$$\eta_a^2 = 2 - \frac{2}{\sqrt{1+k^2 a^2}} \sim k^2 a^2.$$

Write $L_a = \log(1/m_a) \sim \log(1/a)$. Combining these formulas gives

$$\begin{aligned} w_a &\sim K k \sqrt{\ell} a^{3/2} L_a^{1/q} \longrightarrow 0, \\ X_a &\sim K \sqrt{\ell k \mathbb{E}|Z|} a^{1/2} L_a^{1/q}, \\ \log(K/w_a) &\sim \frac{3}{2} L_a. \end{aligned}$$

For the second relation, the angular overlap term in (13) is $O(a^3)$ inside the brackets, whereas D_a is asymptotic to a positive constant times a . It follows that, for every real γ ,

$$\frac{X_a}{K^{2/3} w_a^{1/3} (1 + \log(K/w_a))^\gamma} \sim \ell^{1/3} k^{1/6} \sqrt{\mathbb{E}|Z|} (3/2)^{-\gamma} L_a^{2/(3q) - \gamma}. \quad (14)$$

This proves both the divergence for every smaller logarithmic power and the finite positive limiting ratio at $\gamma = 2/(3q)$. It completes the lower-bound proof of Theorem 1.

4 Upper bound and full Gaussian contrast

Both fixed sources above satisfy manuscript 007, Corollary 3.2: the uniform cube is compact log-concave, and the truncated Gaussian is strongly log-concave with $\kappa = 1$. Their slice variation is finite. Its Corollary 4.1 therefore gives

$$X_\rho(\mu, \nu) \leq C_{q,\rho,d} K^{2/3} w^{1/3} (1 + \log(K/w))^{2/(3q)}, \quad 0 < w \leq K,$$

on $\mathcal{E}_{q,K}$. This is the stated matching upper bound, using the source-dependent potential theorems proved in manuscript 001 v3, Theorems 1.1 and 1.4. The present note does not independently

revalidate those theorems. In particular, subgaussian target tails ($q = 2$) have a necessary logarithmic power $1/3$ for either hard-boundary source.

The conclusion differs for the full standard Gaussian γ_d . Use target moment order $p = 3$, independently of the stretched-exponential parameter $q > 0$. Since

$$s^3 \leq C_q e^{s^q}, \quad C_q = \sup_{s \geq 0} s^3 e^{-s^q} = (3/(qe))^{3/q},$$

the class assumption bounds the sum of third moments by $4C_q K^3$. Thus the sum M of their third-moment roots satisfies

$$M \leq 2^{2/3} (4C_q K^3)^{1/3}.$$

Manuscript 001 v3, Theorem 1.6, applied with moment order three, gives the pure bound

$$\|T_\mu - T_\nu\|_{L^2(\gamma_d)} \leq C_{d,q} K^{2/3} W_2(\mu, \nu)^{1/3} \quad ((\mu, \nu) \in \mathcal{E}_{q,K}),$$

with no logarithmic factor. Its one-third exponent is sharp for $d \geq 2$: the three-atom bounded-target examples of Theorem 1.3 in that manuscript can be scaled into the ball of radius $K(\log 2)^{1/q}$, which places their combined exponential moment at most four. Thus a logarithmic loss is necessary for the fixed hard-boundary strongly log-concave example but absent for the fixed full Gaussian.

Remark 3 (Dimension and other exponential bounds). The lower constructions require $d \geq 2$. In dimension one, an atomless source has the exact monotone-quantile identity $X_\rho = W_2$. The number four in the exponential-moment class is only a convenient normalization. For any fixed $B > 2$, replace $R = K[\log(1/m)]^{1/q}$ by $R = K[\log(c/m)]^{1/q}$ with $c = (B - 2)/2$, for sufficiently small $m < c$. The combined exponential moment becomes $B - 2m \leq B$, and the same asymptotics and logarithmic exponent hold. At $B = 2$ the class contains only the pair (δ_0, δ_0) .

References

- [1] mxym/math research collection, *Sharp Brenier stability under target moment bounds*, version 3, 7 October 2026, manuscript 001, mxym/math. Theorem 1.7 and Section 11 give the cube-truncated Gaussian two-atom family; Section 12.2 gives the exact uniform-cube family. Theorem 1.6 gives the full-Gaussian finite-moment one-third bound; Theorem 1.3 gives its bounded-target sharpness. Theorems 1.1 and 1.4 prove the source-dependent potential estimates used for the upper bounds. Audited source blob 0007edf5e28e4b4b0d34b2043cffd6e13e9f5aed. Pinned source: <https://github.com/mxym/math/blob/0bd6c682edbe6f15001ddc2d73aac3c84ac979a4/preprints/001-strongly-log-concave-brenier/v3/manuscript.tex>.
- [2] mxym/math research collection, *Tail-sensitive convex-gradient interpolation and Brenier stability beyond bounded targets*, version 1, 7 October 2026, manuscript 007, mxym/math. Theorem 3.1, Corollary 3.2, and Corollary 4.1. Audited source blob 2d6f87539f93f87b33e2484334542ab924143057. Pinned source: <https://github.com/mxym/math/blob/0bd6c682edbe6f15001ddc2d73aac3c84ac979a4/preprints/007-tail-brenier-stability/v1/main.tex>.

Both sources are read at repository commit 0bd6c682edbe6f15001ddc2d73aac3c84ac979a4. The note records a new combination of their explicit results; it makes no claim of novelty relative to the literature.